

NAVIN BHATTARAI

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RESEARCH INTERESTS

Thermodynamics & Heat Transfer, Fluid Mechanics, Computational Fluid Dynamics (CFD), Topology Optimization, Sustainable Energy Systems, Thermal Management

EDUCATION

B.S. in Mechanical Engineering

Expected Dec. 2026

Washington University in St. Louis, Missouri

GPA: 3.98/4.00

- Honors: *Lee Hunter Scholarship* (\$3,000/year)
- Relevant Coursework: Heat Transfer, Computational Fluid Dynamics I, Thermo-Fluid System Design (Renewable Energy), Fluid Mechanics, Thermodynamics, Control Systems, Solid Mechanics

B.A. in Physics

Degree Awarded May 2026

Truman State University, Kirksville, Missouri

GPA: 4.00/4.00

- Completed via the Truman State–Washington University 3+2 Dual-Degree Engineering Program (resident at Truman State: August 2022 – May 2025)
- Honors: *Summa Cum Laude*, *Sigma Pi Sigma*, *International President's Honorary Scholarship* (\$7,500/yr), *7+ Merit-Based Scholarships* (\$19,500 total)

RELEVANT RESEARCH EXPERIENCE

Undergraduate Researcher, SURGE Program

May 2026 – Present

PI: Dr. Xianglin Li, Washington University in St. Louis

St. Louis, MO

Project: *Flow Channel Optimization for Heat and Mass Transfer*

- Conducting CFD research on flow-channel optimization for battery cold-plate thermal management, targeting reduced pressure drop while maintaining flow uniformity
- Developing OpenFOAM workflows spanning mesh generation (blockMesh, snappyHexMesh), solver configuration, turbulence modeling, and ParaView post-processing
- Implemented adjoint-based topology optimization, achieving 70+ stable design iterations; extending the method now to handle more complex design problems
- Quantified up to 15.7% pressure-drop reduction via corner-curvature (fillet) optimization of sharp channel turns; curvature proved beneficial above $Re \approx 3000$
- Validated baseline CFD methodology against Ghia et al.'s (1982) lid-driven cavity benchmark
- Executed OpenFOAM simulations with over 72,000 computational cells across 60 CPU cores using shell-scripted workflows

Research Assistant

Feb. 2024 – Aug. 2025

PI: Dr. Rasanjali Jayathissa, Truman State University

Kirksville, MO

Project: *Dye-Sensitized Solar Cells (DSSCs) Using Natural-Dye Sensitizers*

- Investigated beetroot- and sumac-derived natural dyes as sustainable alternatives to conventional metal-complex sensitizers in DSSCs
- Designed experiments for natural-dye extraction and characterization using UV-Vis, Fourier transform infrared (FTIR), and electrochemical impedance (EIS) spectroscopy
- Fabricated and tested 30 laboratory-scale DSSCs using solar-simulator I–V measurements; the best-performing sumac-based cell achieved 0.099% efficiency and a 55.2% fill factor
- Linked higher photocurrent in sumac-based cells to stronger light absorption, and higher open-circuit voltage in beetroot-based cells to lower recombination
- Presented findings at the 2025 APS Prairie Section Meeting

RELEVANT PRESENTATIONS

- **Bhattacharai, Navin.** “Harnessing the power of natural dyes for clean energy: Dye-Sensitized Solar Cells based on plant dyes.” *APS Prairie Section Meeting*, Feb. 21, 2025.

RELEVANT TEACHING EXPERIENCE

Undergraduate Course Assistant

Aug. 2025 – Present

McKelvey School of Engineering, Washington University in St. Louis

St. Louis, MO

- Support instruction in Statics, Mechanics of Materials, Solid Mechanics, and Thermodynamics for 80–150 students per course
- Conduct weekly office hours and individualized problem-solving support; led J-Term instructional sessions in Statics and Mechanics of Materials for a class of approximately 15 students

Academic Peer Tutor

Aug. 2023 – May 2025

Center of Academic Excellence, Truman State University

Kirksville, MO

- Tutored Calculus I–III and introductory Physics I–II through individual and small-group sessions
- Provided in-class embedded tutoring during Fall 2024, reinforcing course material and problem-solving

ENGINEERING & TECHNICAL EXPERIENCE

Co-Lead

Aug. 2025 – Present

WashU Design/Build/Fly, Structures Sub-Team

St. Louis, MO

- Design aircraft structural components and subassemblies in SOLIDWORKS for manufacturability, fabrication, and system integration
- Designed and fabricated a motor test stand for full-throttle thrust and power testing, using stress analysis to balance structural rigidity, portability, and safety

Level 1 Technician

Feb. 2026 – Present

Student Technology Services, Washington University in St. Louis

St. Louis, MO

- Diagnose and resolve hardware, software, network, and other technical issues using systematic troubleshooting
- Support students and faculty via ServiceNow ticketing, personally resolving 6–10 of the team's ~40 weekly tickets

Building Operations Manager

Aug. 2022 – Aug. 2025

Student Union and Involvement Services, Truman State University

Kirksville, MO

- Coordinated 100+ events for 45+ campus organizations under tight turnarounds, managing logistics, event planning, and safety compliance
- Trained and supervised 6+ assistants, enforcing facility standards and operational procedures

HONORS & AWARDS

- Summa Cum Laude, Truman State University, 2026
- Lee Hunter Scholarship, Washington University in St. Louis
- Sigma Pi Sigma, Physics Honor Society
- International President's Honorary Scholarship
- 7+ Merit-Based Academic Scholarships

SKILLS

Simulation & Optimization: OpenFOAM (blockMesh, snappyHexMesh), SU2, COMSOL Multiphysics, ParaView, Topology Optimization (MMA, adjoint methods)

Programming: Python, C++, MATLAB, Mathematica, R, $\text{\LaTeX}$

Mechanical Design: SOLIDWORKS, AutoCAD, Onshape, Finite Element Analysis (FEA), Technical Drawing, DFM/DFA, Tolerance Analysis

Experimental Methods: UV-Vis Spectroscopy, FTIR, Electrochemical Impedance Spectroscopy, Solar Simulator I–V Testing, Prototype Fabrication, Data Analysis

Manufacturing: Machining, Soldering, FDM 3D Printing, CAM / Conversational Programming (ProtoTRAK)